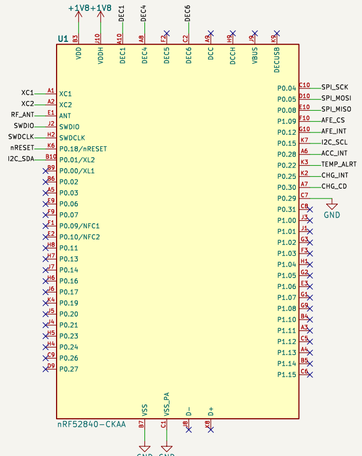
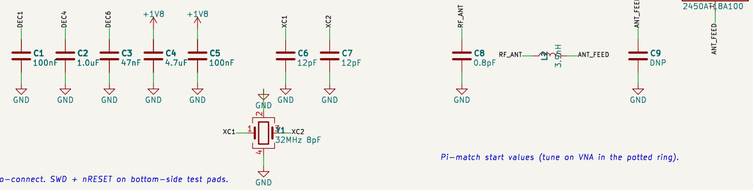
1 BLE SoC + RF

nRF52840-CKAA WLCSP, normal-voltage mode (VDD = VDDH = 1.8 V, internal LDOs)



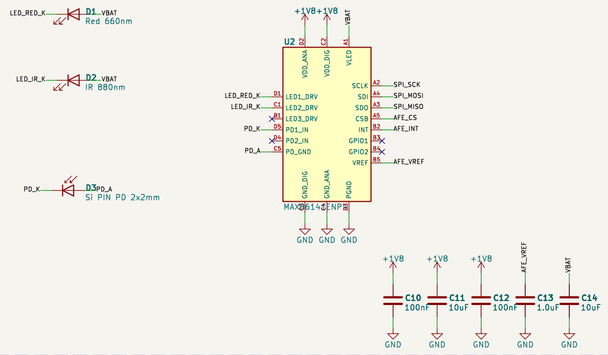


Unused GPIO: no-connect. SWD + nRESET on bottom-side test pads.

Pi-match start values (tune on VNA in the potted ring).

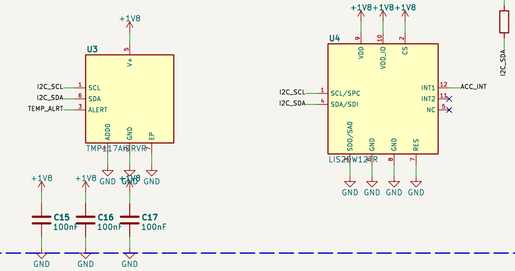
2 Optical front-end

MAX86141 WLP-20, LEDs from VBAT, VDD 1.8 V



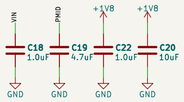
3 Sensors + I2C

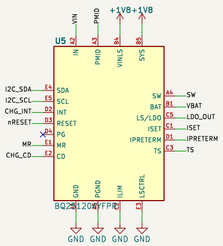
TMP117 (0x48), LIS2DW12 (0x18): 4.7k pull-ups to 1.8 V

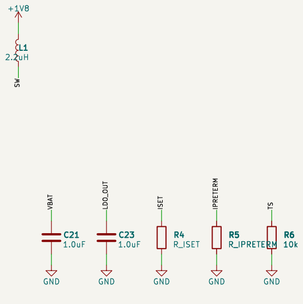


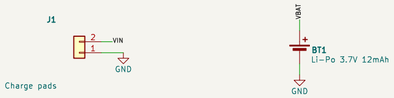
4 Charger, power path, battery, test pads

BQ25120A: buck SYS output set to 1.8 V (I2C), LS/LDO disabled (LSCTRL low), ILIM to GND = internal default









Power flags





R4/R5 values per TI BQ25120A data sheet (ISET, IPRETERM); charge current also set over I2C (12 mAh cell -> about 10 mA).

Rev B: manufacturer pin maps, full support circuitry, one 1.8 V rail
Vishwakarma Awards 2026-27 VK26-AU-FD-0015
Team Ring of Care – Rajalakshmi Engineering College
Sheet: /
File: NadiSense_ring_v2_revB.kicad_sch

Title: NadiSense ring v2 – system schematic

Size: A2	Date: 2026-10-28	**Rev: B**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 1/1